

Introduction

Project title: AI BlogNest API

Team ID : SWTID-2026-5758

Team Size : 4

Team Leader : Varshini.M

Team member : Gladys nancy.A

Team member : Gowri shankari.K

Team member : Gomathi.k

1.Project Overview

AI BlogNest is an AI-powered blog creation and management web application that helps users create, edit, publish, and manage blog posts easily. It uses Artificial Intelligence (AI) to generate blog content based on the topics provided by users.

The main aim of this project is to reduce the time and effort required for writing blogs and make content creation easier and more efficient.

2.Project Objectives

- 1.To develop an AI-based blogging platform for creating and managing blog posts easily.
- 2.To generate blog content using Artificial Intelligence based on the topic or keywords provided by the user.
- 3.To reduce the time and effort required for manual blog writing.
- 4.To provide a simple and user-friendly interface for creating, viewing,

editing, and deleting blogs.

5.To securely store user and blog information using a database.

6.To provide REST APIs for communication between the frontend, backend, database, and AI service.

7.To help users improve their content creation process by providing AI-assisted ideas and content.

8.To provide an organized platform where users can manage all their blogs in one place.

3.Target users

The main target users of AI BlogNest are people who want to create and manage blog content easily with the help of Artificial Intelligence.

1.Students – To create educational blogs, assignments, and project-related content.

2.Bloggers – To generate ideas and create blog posts quickly.

3.Content Writers – To get AI assistance for drafting and improving content.

4.Teachers and Educators – To create educational articles and learning materials.

5.Small Business Owners – To create promotional and informational blog content.

6.Beginners – Users who want to start blogging without advanced writing skills.

4.Pre-requisite & knowledge

requirements

Pre-Requisites

Before developing the AI BlogNest project, the following are required:

1. A computer or laptop with a stable internet connection.
2. VS Code or any suitable code editor.
3. Node.js and npm installed for backend development.
4. A database such as MongoDB for storing user and blog information.
5. An AI API key for AI-based blog content generation.
6. A modern web browser such as Chrome, Edge, or Firefox.
7. Postman for testing the REST APIs.

Knowledge Requirements

The developer should have basic knowledge of:

HTML, CSS, and JavaScript – for frontend development.

Node.js and Express.js – for creating the backend and REST APIs.

MongoDB – for database operations.

REST API concepts – for communication between frontend and backend.

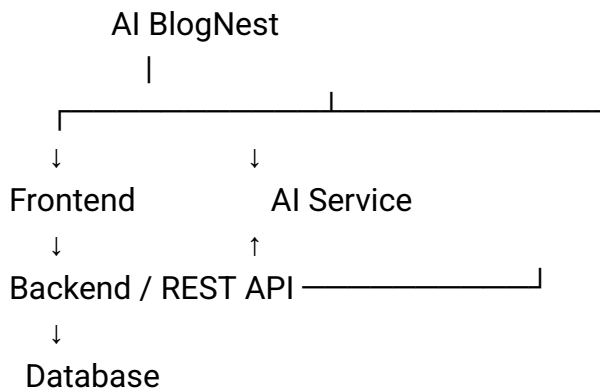
JSON – for transferring data through APIs.

AI API integration – for generating blog content.

Basic authentication concepts – for user registration and login.

CRUD operations – Create, Read, Update, and Delete blog posts.

5. Project architecture and tech stack



1. Frontend Layer

The frontend provides the user interface.

Responsibilities:

- User registration and login

- Dashboard

- Create blog

- AI blog generation

- View, edit, and delete blogs

Technologies: HTML, CSS, JavaScript

2. Backend Layer

The backend processes requests from the frontend and communicates with the database and AI service.

Responsibilities:

User authentication
Blog CRUD operations
REST API handling
AI API integration
Data validation and error handling
Technologies: Node.js, Express.js

6. Project setup and configuration

The following software is required to set up the AI BlogNest project:

- VS Code – Code editor
- Node.js & npm – Backend development
- MongoDB – Database
- Postman – API testing
- Web Browser – To run the application
- Git/GitHub – Version control

```
mkdir AI-BlogNest
cd AI-BlogNest
npm init -y
npm install express mongoose cors dotenv bcryptjs jsonwebtoken
```

These packages are used for:

- Express.js – Creating REST APIs
- Mongoose – Connecting Node.js with MongoDB
- CORS – Allowing frontend-backend communication
- dotenv – Managing environment variables
- bcryptjs – Password encryption
- jsonwebtoken – User authentication

7. Features&api endpoints

Main Features

1.User Registration

New users can create an account.
User details are stored in the database.

2.User Login

Registered users can securely log in.
Authentication is used to protect user-specific features.

3.AI Blog Generation

User enters a topic or prompt.
AI generates suitable blog content automatically.

4.Create Blog

Users can create and save their own blog posts.

5.View Blogs

Users can view available blogs and their own posts.

6.Edit Blog

Users can modify the title or content of an existing blog.

7.Delete Blog

Users can remove unwanted blog posts.

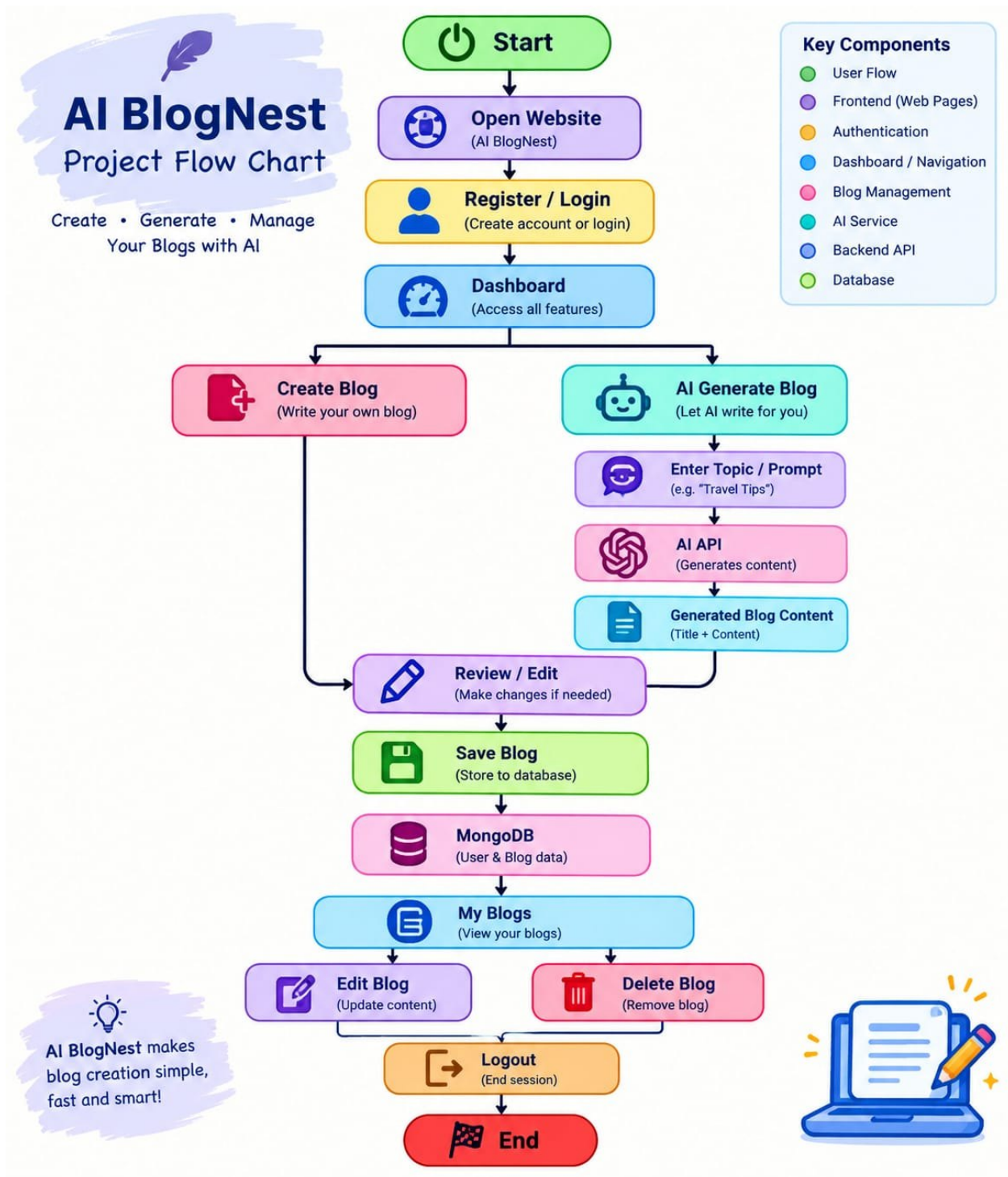
8.Dashboard

Provides access to blog creation, AI generation, and blog management features.

9.Database Management

User and blog information is stored and retrieved from MongoDB.

8. Project flowchart



9. Output

The screenshot displays the Visual Studio Code interface with a REST client request and its output.

REST Client Request:

- Method: POST
- URL: `http://localhost:5000/api/ai/generate-blog`
- Body (JSON):

```
{  "topic": "React js",  "category": "Technical"}
```

Response:

- Status: 500 Internal Server Error
- Size: 164 Bytes
- Time: 82 ms
- Response Body (JSON):

```
{  "message": "[GoogleGenerativeAI Error]: Error fetching from https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent: fetch failed"}
```

Terminal Output:

```
s. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\ELCOT\Downloads\Code Files-20260924T113331Z-1-001\ Code Files> npm install

up to date in 867ms

26 packages are looking for funding
  run `npm fund` for details

PS C:\Users\ELCOT\Downloads\Code Files-20260924T113331Z-1-001\ Code Files> npm start
```

Explorer

Open Editors

New Request

localhost:5000/api/auth/r...

localhost:5000/api/auth/r...

.env Code Files

Code Files-20260924T113331Z-1-001

Code Files

node_modules

src

config

controllers

middleware

models

routes

services

app.js

server.js

.env

.gitignore

package-lock.json

package.json

README.md

thunder-client-ai-blognest-api.postman_c...

thunder-client-env.json

package-lock.json

Outline

New Request

POST http://localhost:5000/api/ai/generate-blog

Send

Query Headers Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content

```
1 {
2   "topic": "React js",
3   "category": "Technical"
4 }
```

Format

Status: 500 Internal Server Error Size: 164 Bytes Time: 82 ms

Response Headers Cookies Results Docs

```
1 {
2   "message": "[GoogleGenerativeAI Error]: Error fetchi
   from https://generativelanguage.googleapis.com
/v1beta/models/gemini-2.5-flash:generateContent:
   fetch failed"
3 }
```

Response Chart

Problems Output Debug Console Terminal Ports

run `npm fund` for details

PS C:\Users\ELCOT\Downloads\Code Files-20260924T113331Z-1-001\ Code Files> npm start

```
> ai-blognest-api@1.0.0 start
> node src/server.js
```

Server running on port 5000

MongoDB connected: localhost

POST /api/auth/register 201 225.313 ms - 126

Explorer

Open Editors

New Request

localhost:5000/api/auth/r...

localhost:5000/api/auth/r...

.env Code Files

Code Files-20260924T113331Z-1-001

Code Files

node_modules

src

config

controllers

middleware

models

routes

services

app.js

server.js

.env

.gitignore

package-lock.json

package.json

README.md

thunder-client-ai-blognest-api.postman_c...

thunder-client-env.json

package-lock.json

Outline

Timeline

New Request

POST http://localhost:5000/api/ai/generate-blog

Send

Query Headers Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content

```
1 {
2   "topic": "React js",
3   "category": "Technical"
4 }
```

Format

Status: 500 Internal Server Error Size: 164 Bytes Time: 82 ms

Response Headers Cookies Results Docs

```
1 {
2   "message": "[GoogleGenerativeAI Error]: Error
   from https://generativelanguage.googleapis
/v1beta/models/gemini-2.5-flash:generateCo
   fetch failed"
3 }
```

Response C

Problems Output Debug Console Terminal Ports

MongoDB connected: localhost

POST /api/auth/register 201 225.313 ms - 126

POST /api/auth/login 200 168.867 ms - 183

POST /api/blogs 201 50.310 ms - 245

SyntaxError: Unexpected token '...', ""{\n \to"... is not valid JSON

at JSON.parse (<anonymous>)

at createStrictSyntaxError (C:\Users\ELCOT\Downloads\Code Files-20260924T113331Z-1-001\ Code Files\node_modules\body-p

arser\lib\types\json.js:165:10)

at parse (C:\Users\ELCOT\Downloads\Code Files-20260924T113331Z-1-001\ Code Files\node_modules\body-parser\lib\types\js

10.Conclusion

AI BlogNest is an AI-powered web application designed to make blog creation and management simple and efficient. It allows users to create, edit, view, and delete blogs while using AI to generate content based on a given topic.

The project combines frontend technologies, backend REST APIs, MongoDB, and AI services to provide an integrated blogging platform. It reduces the time and effort required for content creation and provides a user-friendly experience.

Overall, AI BlogNest demonstrates how Artificial Intelligence can be integrated with web applications to improve and simplify digital content creation.